# What Can IRT Person-Fit Statistics Reveal About SFT-Stage Benchmark Contamination?

**Anonymous ACL submission**

## Abstract

Benchmark contamination in Large Language Models (LLMs) undermines evaluation validity, yet most detection methods require privileged access to training data, model internals, or output probabilities. We present the first controlled characterization of IRT person-fit statistics for detecting SFT-stage contamination, requiring only binary response patterns and externally calibrated 4PL item parameters; diagnostic $\theta$-anchoring additionally requires a clean baseline. A contaminated LLM produces aberrant response patterns analogous to a test-taker with item preknowledge. Using $\ell_z^*$ in a matched-difference design ($\Delta\ell_z^*$) across three 7–8B model families ($n$=10 clean baseline seeds per model family) on MMLU (14,042 items) and ARC-Challenge (295 items), we identify three behavior boundaries: (1) model-specific $\theta$-shift dynamics, where contamination-induced ability inflation absorbs the person-fit signal; $\theta$-anchoring is necessary for reliable detection of high-ability models at high contamination (2.4$\times$ amplification, inter-seed CV reduced from 60% to 19%); (2) a detection-diagnosis tradeoff where simpler statistics yield stronger detection but lack $\ell_z^*$'s diagnostic capabilities (seen/unseen decomposition, mechanistic $\theta$-shift analysis); and (3, exploratory) ability-dependent non-monotonic dose-response under free-$\theta$ estimation, driven by $\theta$-shift absorption; $\theta$-anchored analysis reveals stable detection with a strong positive dose-response at higher contamination levels. These findings establish when and why binary response patterns carry contamination information beyond aggregate accuracy.

## 1 Introduction

Benchmark contamination, the unintended inclusion of evaluation data in training corpora, poses a growing threat to the integrity of large language model (LLM) evaluation (Dodge et al., 2021; Magar and Schwartz, 2022; Sainz et al., 2023). As training corpora and public benchmarks both expand, the probability of overlap increases structurally (Jacovi et al., 2023; Chang et al., 2024). When contamination occurs, benchmark scores no longer reflect genuine capability, undermining the primary mechanism by which the community compares and selects models.

A range of detection methods have been proposed, but all require some form of privileged access beyond the binary response record. N-gram overlap methods need the training corpus (Dodge et al., 2021; Magar and Schwartz, 2022). Membership inference approaches require token-level probabilities (Shi et al., 2024; Deng et al., 2024). Probing methods rely on crafted prompts (Golchin and Surdeanu, 2024a; Oren et al., 2024). Performance-based methods require multiple test variants or output logits (Dekoninck et al., 2024; Zawalski et al., 2026). These methods form a detection frontier ordered by information access. No existing method operates using only binary response patterns, without model weights, gradients, logits, or training data (though, as we show, effective detection additionally requires externally calibrated item parameters).

In this work, we focus on SFT-stage contamination, where benchmark answers are explicitly included in fine-tuning data, for three reasons. First, controlled experiments require known ground-truth contamination sets, achievable through fine-tuning with specific benchmark items. Second, SFT-stage contamination is increasingly relevant as fine-tuning-as-a-service platforms proliferate, creating new pathways for benchmark data to enter training mixtures. Third, SFT provides a methodologically clean setting to characterize detection boundaries before extending to noisier pretraining scenarios.

The need for detection from binary response patterns grows more pressing as models evolve. Wang et al. (2025) show that all ten log-probability-based methods they evaluate perform at chance level on

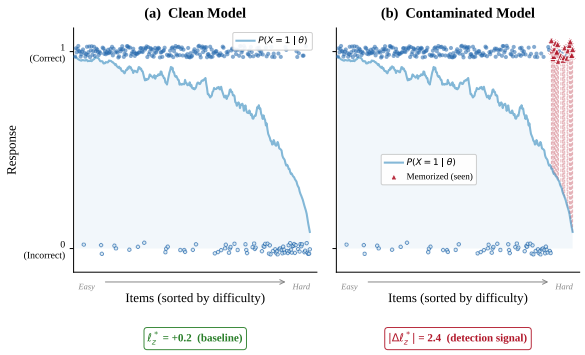



Figure 1: A clean model's responses track the IRT-predicted probability curve (left); a contaminated model shows anomalous successes on hard items (right, red diamonds), producing a large $|\Delta\ell_z^*|$ signal.

models trained with reinforcement learning (RL), because RL reshapes internal token distributions while preserving binary correctness patterns. This motivates a fundamental question: *what can binary response patterns reveal about contamination, and under what conditions?*

We observe that a contaminated LLM is functionally analogous to a test-taker with item preknowledge: both produce response patterns in which difficult items are answered correctly through memorization rather than genuine ability. This connection points to person-fit statistics, developed over four decades for cheating detection in educational testing (Drasgow et al., 1985; Meijer and Sijtsma, 2001; Karabatsos, 2003; Sinharay, 2017). The standardized statistic $\ell_z^*$ (Snijders, 2001) measures how well a response pattern conforms to Item Response Theory (IRT) model predictions. Figure 1 illustrates this analogy. Despite the natural correspondence, no prior work has applied person-fit statistics to LLM benchmark contamination.

Rather than proposing $\ell_z^*$ as a ready-to-deploy detection tool, we characterize the behavior boundaries of IRT person-fit statistics for LLM contamination detection. Our framework operates at three layers: (i) pre-calibrated 4PL item parameters from the PSN-IRT item bank (Zhou et al., 2026), an external resource calibrated from large-scale leaderboard data; (ii) binary response patterns obtained from the target model without access to weights or logits; and (iii) retrospective statistical inference requiring no model modification. Using controlled contamination experiments across three 7–8B model families (Qwen2.5-7B, Llama-3.1-8B, Mistral-7B; $n$=10 clean seeds each) with PSN-IRT parameters, we identify three IRT-specific findings on MMLU (Hendrycks et al., 2021) and ARC-Challenge:

1. **Model-specific $\theta$-shift dynamics.** Contamination inflates ability estimates ($\hat{\theta}$), absorbing the person-fit signal; $\theta$-anchoring amplifies Qwen's multi-seed $|d|$ by 2.4× at 50% contamination while reducing inter-seed CV from 60% to 19%, establishing $\theta$-anchoring as a necessary condition for reliable detection of high-ability models at high contamination. ARC-C reverses through a ceiling effect.
2. **Detection-diagnosis tradeoff.** Simpler statistics yield stronger detection: accuracy $z$-scores produce $|d|$ values several-fold to over an order of magnitude larger than person-fit statistics, and HT generally exceeds $\ell_z^*$. Yet the diagnostic power reverses: only $\ell_z^*$ supports seen/unseen decomposition, mechanistic $\theta$-shift analysis, and sensitivity to response-pattern structure (57/60 vs. 4/60 flagged cells on real-world data).
3. **Non-monotonic dose-response under free-$\theta$ (exploratory).** The Qwen dose-response ($|d|$: 4.15 → 5.36 → 7.44 → 5.08 at 5%/15%/25%/50%; 5% is single-seed, remainder are multi-seed means) shows non-monotonicity; $\theta$-anchoring eliminates the pattern and reveals strong positive dose-response ($|d|$=12.32 at 50%). The non-monotonicity is a $\theta$-estimation artifact, not a detection boundary.

Our contributions are threefold: (1) We provide the first empirical characterization of IRT person-fit behavior for LLM contamination, quantifying theoretically predicted phenomena ($\theta$-shift absorption, detection-diagnosis tradeoff) whose magnitudes and boundary conditions are not derivable from IRT theory alone. (2) We establish that $\Delta\ell_z^*$ reliably detects SFT-stage contamination across three model families (all bootstrap 95% CIs exclude zero, $n$=10). (3) We document the detection-diagnosis tradeoff, clarifying what binary response patterns can and cannot reveal. Findings 1–2 are confirmatory (pre-planned 15%/50% × 3 families); Finding 3 is exploratory (5%/25% added post-hoc).

## 2 Related Work

**Benchmark contamination detection.** Contamination detection methods span a wide range of information requirements. Corpus-search methods match benchmark items against pretraining data (Dodge et al., 2021; Magar and Schwartz, 2022)

but require full corpus access. Membership inference attacks use token-level log-probabilities (Shi et al., 2024; Deng et al., 2024); generation-based probes elicit benchmark content via prompting (Golchin and Surdeanu, 2024a,b; Oren et al., 2024); performance-based methods compare accuracy across target and reference sets (Dekoninck et al., 2024; Zhang et al., 2024; Zawalski et al., 2026). Several works address contamination measurement and mitigation (Zhou et al., 2023; Sainz et al., 2023; Jacovi et al., 2023), while Bordt et al. (2025) show contamination can be "forgotten" under continued training, Han et al. (2026) extend contamination quantification to psychometric evaluations, and Kattamuri et al. (2025) use activation patterns for mechanistic detection. Fu et al. (2025) systematically evaluate detection method assumptions, finding degradation under realistic conditions.

A critical limitation unites token-level methods: Wang et al. (2025) show RL post-training renders MIA-based detectors no better than random, because RL reshapes token distributions while preserving binary correctness patterns. Tao et al. (2026) and Zhang et al. (2026) propose RL-specific solutions but require generation-level access. Our approach requires only binary response patterns and externally calibrated item parameters from PSN-IRT (Zhou et al., 2026).

**IRT for LLM evaluation.** IRT has seen growing adoption in ML evaluation: Martínez-Plumed et al. (2019) applied IRT to analyze classifiers at the instance level; Lalor et al. (2019) estimated IRT parameters from DNN response patterns; Rodriguez et al. (2021) showed IRT-informed item weighting improves rankings; Polo et al. (2024) reduced benchmark sizes by 100× via IRT-based item selection; and Li et al. (2025) developed adaptive testing for efficient LLM evaluation. Most directly relevant, Zhou et al. (2026) calibrated 4PL item parameters across 11 benchmarks (41,871 items), providing the item bank our framework builds on. None of these works apply *person-fit* statistics for contamination detection.

**Person-fit statistics in psychometrics.** Person-fit analysis has a long history in educational measurement (see Embretson and Reise, 2000). Drasgow et al. (1985) introduced $\ell_z$, a standardized log-likelihood residual; Snijders (2001) derived the corrected $\ell_z^*$ accounting for ability estimation uncertainty. Karabatsos (2003) compared 36 person-fit statistics and found $\ell_z$ among the most powerful for detecting cheating; we select $\ell_z^*$ for its theoretical grounding and HT (Van der Flier, 1982) as a nonparametric complement. Sinharay (2017) showed $\ell_z^*$ is particularly effective when compromised items are known, analogous to our seen/unseen decomposition. Our work transfers this psychometric toolkit to LLM evaluation, treating each model as an examinee whose response pattern may reveal memorization.

## 3 Method

### 3.1 IRT Person-Fit Framework

We adopt the four-parameter logistic (4PL) IRT model to characterize contamination-induced aberrance in LLM response patterns. The 4PL model defines the probability that a model with ability $\theta$ answers item $i$ correctly:

$$P_i(\theta) = c_i + (d_i - c_i)\,\frac{1}{1 + e^{-a_i(\theta - b_i)}} \qquad (1)$$

where $a_i$ is discrimination, $b_i$ difficulty, $c_i$ the lower asymptote (guessing), and $d_i$ the upper asymptote (slip). Item parameters are pre-calibrated by PSN-IRT (Zhou et al., 2026) from a large-scale leaderboard dataset; we use these external parameters rather than re-estimating from potentially contaminated data, since our MML-EM refit shows discrimination recovery degrades to $r_a$=0.127 on contaminated leaderboard data.

Given item parameters and a binary response vector $\mathbf{x} = (x_1, \ldots, x_n)$, we compute the person-fit statistic $\ell_z^*$ (Snijders, 2001), a standardized log-likelihood residual quantifying deviation between observed and expected response patterns. The Snijders correction accounts for estimation uncertainty in $\hat{\theta}_{\text{MLE}}$, so that $\ell_z^* \sim \mathcal{N}(0,1)$ under the null (Drasgow et al., 1985; Snijders, 2001); Magis et al. (2012) provide a detailed treatment of how response model selection and ability estimation affect $\ell_z^*$. Large negative $\ell_z^*$ indicates aberrant responding (unexpected failures on easy items or unexpected successes on hard items); large positive values indicate over-consistent patterns.

### 3.2 $\Delta\ell_z^*$ Matched-Difference Design

LLM response patterns systematically violate IRT assumptions of local independence and unidimensionality. In our empirical data, 57 of 60 model-benchmark cells yield $|\ell_z^*| > 1.96$ even absent

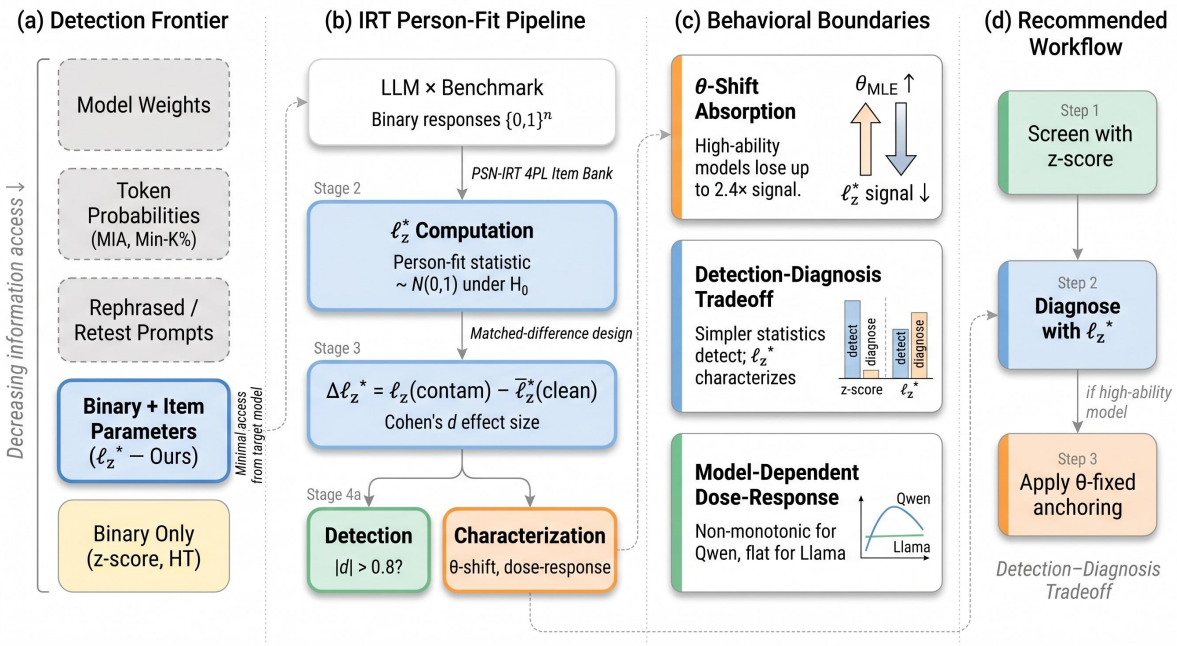



Figure 2: Overview of IRT-FIT: (a) positioning in the detection frontier by information access; (b) pipeline from binary responses through $\ell_z^*$ person-fit analysis to detection and characterization; (c) three behavioral boundaries discovered; (d) recommended practitioner workflow.

contamination, rendering absolute thresholds uninformative. We address this through a matched-difference design.

For controlled experiments with clean baselines, we compute:

$$\Delta\ell_z^* = \ell_{z,\text{contam}}^* - \bar{\ell}_{z,\text{clean}}^* \tag{2}$$

where $\bar{\ell}_{z,\text{clean}}^*$ averages over $n$=10 clean seeds. The shared model misfit cancels, isolating contamination-specific aberrance. We quantify effect sizes as $d = (\ell_{z,\text{contam}}^* - \bar{\ell}_{z,\text{clean}}^*) \,/\, s_{\text{clean}}$, where $s_{\text{clean}}$ is the SD of the $n$=10 clean-seed $\ell_z^*$ values; with a single contaminated observation per condition, $d$ is a standardized departure rather than a two-sample effect size (Hedges' $g = J \cdot d$, $J \approx 0.914$ for $n$=10; Hedges, 1981). We report percentile bootstrap 95% CIs (10,000 resamples) reflecting clean-seed variability only; BCa correction (Appendix H) generally confirms conservative coverage; multi-seed contamination-run variance appears in Appendix L.

For deployment without clean baselines, cross-benchmark comparison provides the reference: the target benchmark's $\ell_z^*$ is compared against benchmarks presumed clean from the same model. Significance thresholds derive from a nonparametric bootstrap null rather than the asymptotic $\mathcal{N}(0,1)$ reference, accommodating residual model misspecification. Simulation analysis (Appendix G.1) confirms that the matched-difference design is robust to local dependence at realistic levels ($d_{\text{ratio}} \geq 0.93$ for $\sigma_\tau \leq 0.5$) and maintains $\geq$98.5% power even under contamination-specific asymmetric dependence.

### 3.3 Experimental Setup

**Models and contamination.** We validate on three 7–8B model families: Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, and Mistral-7B-Instruct-v0.3. Each is fine-tuned with QLoRA (rank = 16, $\alpha = 32$; full hyperparameters in Appendix K) under clean and contaminated conditions at varying levels: at $X$% contamination, $X$% of SFT training examples are benchmark items (questions paired with gold answers in the standard evaluation format), with the remainder drawn from a general instruction-following dataset. All three families are tested at 15% and 50% contamination. Following initial observation of non-monotonic dose-response for Qwen, we added 5% and 25% conditions for Qwen to trace the full dose-response curve. Both conditions were subsequently extended to Llama to test whether the pattern generalizes; Mistral was additionally tested at 5% but not at 25% due to computational constraints. We train $n$=10 independent clean seeds per model to establish baseline

variability and construct bootstrap confidence intervals. Reported CIs reflect variability across $n$ clean baseline seeds; for each model family, selected contamination levels were additionally replicated with $n$=3 independent QLoRA seeds to assess training-run variance: Qwen at 15%/25%/50%, Llama at 5%/15%/25%/50%, and Mistral at 5%/15%/50% (Appendix L).

**Benchmarks.** We evaluate on MMLU (Hendrycks et al., 2021) (14,042 items) and ARC-Challenge (Clark et al., 2018) (295 items), spanning two orders of magnitude in benchmark size. All 4PL item parameters are from Zhou et al. (2026). ARC-Challenge experiments were conducted on Qwen only, as the primary evaluation of cross-benchmark generalizability; extending to additional families was constrained by computational budget.

**Post-RL extension.** To probe behavior boundaries beyond SFT, we apply Group Relative Policy Optimization (GRPO; Shao et al., 2024) to one contaminated Qwen model (15% MMLU) for 1,500 steps, evaluating at three checkpoints. This extension is $N$=1 (single model family, single contamination level).

**Contamination model.** QLoRA SFT on benchmark items serves as a controlled proxy for SFT-stage contamination, where benchmark data is explicitly included in fine-tuning mixtures, analogous to a test-taker with item preknowledge. We refer to the contaminated benchmark items as *seen items* and the remaining benchmark items not included in training as *unseen items*; this partition enables decomposition of person-fit statistics to assess whether the contamination signal concentrates on memorized items (§4.3). Our findings are scoped to SFT-based contamination and may not generalize to pretraining contamination, which operates through different learning dynamics.

# 4 Results

## 4.1 Detection Signal Across Model Families

Table 1 presents Cohen's $d$ effect sizes for $\ell_z^*$ across three model families and two benchmarks. All single-seed bootstrap CIs exclude zero, confirming that $\Delta\ell_z^*$ reliably separates contaminated from clean models at both contamination levels.

On MMLU, effect sizes range from $|d|$=2.30 (Llama at 50%) to $|d|$=7.44 (Qwen at 25%), with Qwen at 5% ($|d|$=4.15) confirming detection at the lowest contamination level tested. Llama ($|d|$=2.57/2.80/2.56/2.30) and Mistral ($|d|$=3.83/3.37/4.09) show comparable magnitudes. On ARC-C (295 items), Qwen yields $|d|$=3.16 at 15% and $|d|$=5.30 at 50%, demonstrating that detection remains feasible even on small benchmarks when contamination is sufficiently concentrated.

Table 1: Cohen's $d$ for $\Delta\ell_z^*$ across model families ($n$=10 clean seeds). Single-seed conditions: percentile bootstrap 95% CI (10,000 resamples). †Multi-seed conditions: mean ± SD ($n$=3 independent runs). All single-seed CIs exclude zero.

| Model | Bench. | $d$ (5%) [CI] | $d$ (15%) [CI] | $d$ (25%) [CI] | $d$ (50%) [CI] |
|---|---|---|---|---|---|
| Qwen† | MMLU | −4.15 [−9.9, −2.7] | −5.36 (±1.10) | −7.44 (±1.86) | −5.08 (±3.02) |
| Llama† | MMLU | −2.57 (±0.99) | −2.80 (±0.60) | −2.56 (±0.52) | −2.30 (±0.30) |
| Mistral† | MMLU | −3.83* (±1.42) | −3.37 (±0.68) | — | −4.09 (±0.53) |
| Qwen† | ARC-C | — | +3.16 [2.0, 9.5] | — | +5.30 [3.6, 15.2] |

†Multi-seed conditions: means ($n$=3 independent runs) shown as ±SD; remaining conditions are single-seed with bootstrap CIs.
*$n$=2 seeds. Per-seed values in Appendix L.

Clean-model $\ell_z^*$ baselines diverge substantially across families: Qwen +8.68 (SD=0.74), Mistral +4.02 (SD=2.05), Llama +0.26 (SD=1.54), and ARC-C Qwen −5.42 (SD=0.16); this divergence reflects difficulty-ordering consistency rather than accuracy level (Appendix D). Absolute $\ell_z^*$ thresholds would produce contradictory results: Qwen's contaminated $\ell_z^*$ at 15% (+3.8) is still positive. The $\Delta\ell_z^*$ design normalizes these divergent baselines: Cohen's $d$ yields comparable effect sizes ($|d|$=2.30–7.44) despite order-of-magnitude differences. Figure 3 provides a visual summary.

## 4.2 Model-Specific $\theta$-Shift Dynamics

The cross-family variation in Table 1 (Qwen's $|d|$ declining from 25% to 50% while Llama's remains flat) suggests detection sensitivity depends on more than contamination dose. Contamination raises overall accuracy, which inflates $\hat{\theta}_{\text{MLE}}$; a higher $\hat{\theta}$ "explains away" anomalous successes on hard items, absorbing the contamination signal into the ability estimate. We term this *$\theta$-shift absorption*. The magnitude of $\theta$-shift is comparable across families (Qwen $\Delta\theta$=0.47 at 50%, Llama $\Delta\theta$=0.52), yet their fixed-$\theta$ amplification diverges markedly (Table 2), indicating that $\theta$-shift magnitude alone does not predict fixed-$\theta$ benefit. Figure 4 plots baseline accuracy against Cohen's $|d|$ for all model-benchmark combinations.

$\theta$-shift absorption is the dominant source of detection variability for high-ability models at high contamination: (i) fixed-$\theta$ estimation amplifies

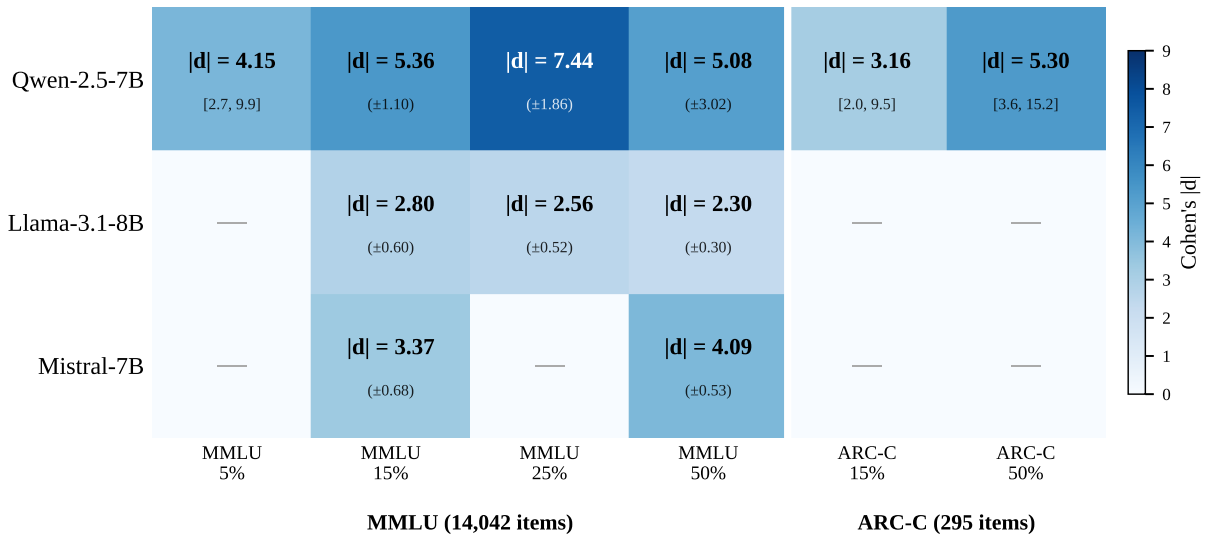



Figure 3: Cohen's $|d|$ heatmap (values from Table 1; multi-seed means where available). Blue intensity reflects effect-size magnitude. Grey cells: conditions not tested.

Qwen's multi-seed $|d|$ by 2.4× at 50% (5.08 → 12.32; Table 2), while Llama shows no amplification (0.4×); (ii) $\theta$-anchoring reduces Qwen's inter-seed CV from 60% to 19%, confirming that training-run variance is driven by $\theta$-shift stochasticity rather than true signal variance; (iii) the cross-model ranking under free-$\theta$ is seed-dependent at 50%, making $\theta$-anchoring a necessary condition for reliable detection of high-ability models at high contamination.

ARC-C reverses this pattern. Despite Qwen's high baseline accuracy on ARC-C (89.7%), contamination produces larger, not smaller, effect sizes ($|d|$=3.16 at 15%, $|d|$=5.30 at 50%). This reversal reflects a ceiling effect: with only 295 items and 89.7% baseline accuracy, the already-correct items offer minimal room for further $\theta$ inflation, so contamination manifests primarily as anomalous successes on the few remaining hard items. The small item count amplifies this concentration, producing a disproportionately large $\ell_z^*$ shift per contaminated item.

The baseline $\ell_z^*$ for ARC-C is negative (−5.42), reflecting the 4PL model's expectation of near-perfect performance; contamination pushes $\ell_z^*$ toward zero, producing the positive Cohen's $d$ values in Table 1.

These findings indicate that detection sensitivity is not a fixed property of $\ell_z^*$ but depends on the interaction between model ability and benchmark characteristics, a consideration absent from standard person-fit applications.

**Fixed-$\theta$ estimation.** Table 2 compares free-$\theta$ and fixed-$\theta$ Cohen's $d$ across all three families. The benefit of fixed-$\theta$ is modulated by baseline $\ell_z^*$ level rather than $\theta$-shift magnitude. Qwen (67.8% accuracy, $\Delta\theta$=0.47, baseline $\ell_z^*$=+8.68) shows 2.4× amplification at 50%: its large baseline departure from the IRT null provides substantial signal that free-$\theta$ absorption erodes, which $\theta$-anchoring recovers (multi-seed mean $\ell_{z\,\text{fixed}}^*$=14.5 vs. clean 7.35). Mistral (52.4%, baseline $\ell_z^*$=+4.02) shows 1.7× at 50%. Llama (59.4%, $\Delta\theta$=0.39 at 15%, 0.52 at 50%) shows 0.4× at 50% despite comparable $\theta$-shift: its near-zero baseline $\ell_z^*$ (+0.26) indicates that the clean response pattern conforms to IRT expectations, leaving no absorbed signal for $\theta$-anchoring to recover. The amplification ratio tracks baseline $|\ell_z^*|$ monotonically (8.68→2.4×; 4.02→1.7×; 0.26→0.4× at 50%), making fixed-$\theta$ a targeted remedy for models with substantial baseline misfit, not a universal one.

### 4.3 Detection-Diagnosis Tradeoff

Having established that $\theta$-shift complicates detection, we ask whether the full 4PL framework is necessary, or whether simpler statistics suffice. Table 3 compares three statistics at decreasing levels of information access: $\ell_z^*$ (requires 4PL parameters $a, b, c, d$), the HT statistic (requires only difficulty $b$), and an accuracy $z$-score (requires no item parameters).

Table 2: Free-$\theta$ vs. fixed-$\theta$ Cohen's $|d|$ on MMLU. Qwen[§]: multi-seed mean ($n$=3); Llama and Mistral: seed-42 single run. Fixed-$\theta$ anchors $\hat{\theta}$ at the clean-model mean, eliminating $\theta$-shift absorption.

| Model | $\lvert d_{\text{free}}\rvert$ | $\lvert d_{\text{fixed}}\rvert$ | Ratio (fixed/free) | Fixed CI |
|---|---|---|---|---|
| *5% contamination* | | | | |
| Qwen | 4.15 | 3.52 | 0.8× | [2.9, 5.7] |
| *15% contamination* | | | | |
| Qwen[§] | 5.36 | 6.85 | 1.3× | (±1.80) |
| Llama | 2.11 | 0.58 | 0.3× | [0, 2.37][†] |
| Mistral[‡] | 2.64 | 2.69 | 1.0× | [1.53, 9.06] |
| *50% contamination* | | | | |
| Qwen[§] | 5.08 | 12.32 | 2.4× | (±2.38) |
| Llama | 2.33 | 0.84 | 0.4× | [0.17, 2.97] |
| Mistral[‡] | 3.50 | 5.80 | 1.7× | [3.87, 18.45] |

[†]CI includes zero.
[§]Multi-seed mean ($n$=3); CI column shows ±SD of fixed-$\theta$ $|d|$.
[‡]Fixed-$\theta$: 6 seeds (4 response vectors from seeds 4–7 not retained in pipeline); free-$\theta$ uses all 10.
$d_{\text{free}}$ CIs omitted; multi-seed means in Table 1.

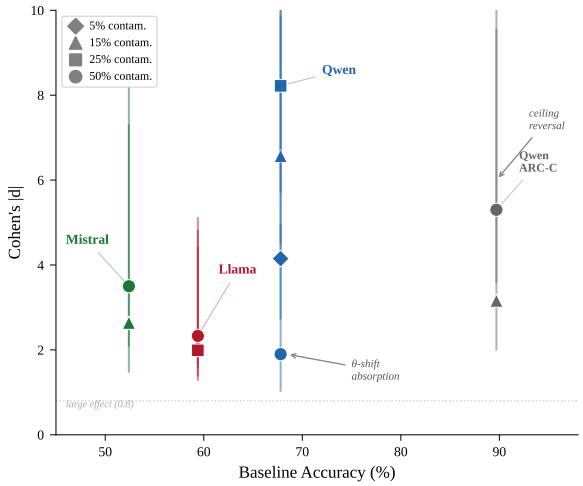



Figure 4: Baseline accuracy vs. Cohen's $|d|$ for all model-benchmark combinations (single-seed values; multi-seed means in Table 1). Diamonds: 5%; triangles: 15%; squares: 25%; circles: 50%. ARC-C reversal reflects a ceiling effect.

The comparison reveals a detection-diagnosis tradeoff. For pure detection, simpler statistics are more powerful: accuracy $z$-scores yield $|d|$ values several-fold to over an order of magnitude larger than person-fit statistics (Table 3), because cross-seed accuracy variance is near-zero in controlled settings. HT generally exceeds $\ell_z^*$ in detection strength, since HT is not attenuated by the Snijders correction for $\theta$ estimation uncertainty. Both person-fit statistics exhibit seed-dependent non-monotonicity under free-$\theta$ estimation (Table 3), confirming that $\theta$-shift absorption affects all statistics conditioned on ability, not $\ell_z^*$ specifically.

Yet the diagnostic power reverses. Full IRT parameterization is unnecessary for mere detection, but the value of $\ell_z^*$ lies in three capabilities that HT and $z$-scores lack: (a) *seen/unseen decomposition*: $\ell_z^*$ can be computed separately on seen and unseen item subsets, though item-level detection is near-random (AUROC $\approx 0.53$; Appendix F), confirming that the contamination signal is diffuse and requires model-level aggregation; (b) *$\theta$-shift mechanism*: the ability parameter exposes why detection sensitivity depends on model ability (§4.2), a mechanistic insight inaccessible to non-parametric statistics; (c) *structural sensitivity*: on real leaderboard data, $\ell_z^*$ flags 57/60 model-benchmark cells as aberrant, evidence of systematic IRT assumption violations that motivates the $\Delta\ell_z^*$ matched-difference design (§3.2; Appendix E). Accuracy $z$-scores flag only 4/60 cells, making them insensitive to the response-pattern structure that $\ell_z^*$ exploits for within-model diagnosis. While such patterns have theoretical antecedents in person-fit literature (St-Onge et al., 2011), the specific magnitude and model-dependence we observe are new empirical findings in the LLM domain; nonparametric alternatives (Tendeiro and Meijer, 2014) may complement $\ell_z^*$ where 4PL calibration is unavailable.

In practice, a two-stage workflow follows: screen using accuracy $z$-scores (Table 3), then apply $\ell_z^*$ to characterize whether the signal reflects $\theta$-shift absorption (indicating fixed-$\theta$ anchoring is needed) or uniform aberrance (suggesting direct memorization).

Within the IRT family, the 4PL specification matters: simulation ablations show a mean +27.3pp power gain over 3PL on MMLU, because 46% of MMLU items have upper asymptote $d < 0.95$, and the 3PL model (which assumes $d$=1) misattributes slip-related response patterns as aberrance (Appendix B).

### 4.4 Ability-Dependent Non-Monotonic Dose-Response

As an exploratory observation, the free-$\theta$ dose-response is non-monotonic for Qwen. Figure 5 plots the dose-response trajectory, combining the single-seed 5% estimate (hollow marker) with multi-seed means at 15%/25%/50% ($|d|$: $4.15 \rightarrow 5.36 \rightarrow 7.44 \rightarrow 5.08$), showing signal peaking at 25% before declining at 50%; the inter-seed variance at 50% is high (CV=60%, range $[1.90, 7.91]$).

This non-monotonicity is a $\theta$-estimation artifact: fixed-$\theta$ analysis eliminates the inverted-U and reveals monotonic dose-response (mean $|d|$: 5.89 at 25% vs. 12.32 at 50%, no seed-level overlap; CV reduced to 19%). Llama shows a flat trajectory

Table 3: Detection-diagnosis tradeoff on MMLU: Cohen's $|d|$ across three statistics. Qwen$^{\dagger}$: multi-seed means ($n$=3); Llama and Mistral: seed-42 single run with bootstrap CIs.

| Model | Statistic | $\|d\|$ (15%) | $\|d\|$ (50%) |
|---|---|---|---|
| Qwen$^{\dagger}$ | $\ell_z^*$ | 5.36 ($\pm$1.10) | 5.08 ($\pm$3.02) |
| | HT | 5.83 ($\pm$1.30) | 5.29 ($\pm$3.41) |
| | Acc. $z$ | 45.0 ($\pm$2.5) | 70.0 ($\pm$1.1) |
| Llama | $\ell_z^*$ | 2.11 [1.4, 4.8] | 2.33 [1.6, 5.1] |
| | HT | 2.45 [1.5, 4.3] | 3.07 [1.9, 5.3] |
| | Acc. $z$ | 22.5 [13.6, 47.0] | 32.8 [19.8, 68.6] |
| Mistral | $\ell_z^*$ | 2.64$^{\ddagger}$ | 3.50 [2.1, 9.4] |
| | HT | 4.36 [1.2, 7.0] | 5.89 [1.9, 9.4] |
| | Acc. $z$ | 7.81 [4.4, 14.4] | 16.6 [9.7, 34.6] |

$^{\dagger}$Per-seed values in Appendix L. Bootstrap CIs: 10,000 resamples over $n$=10 clean seeds. $^{\ddagger}$CI not retained from analysis pipeline; multi-seed SD in Table 1.

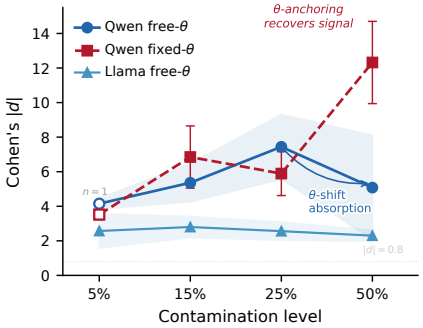



Figure 5: Dose-response trajectories for Cohen's $|d|$ on MMLU. Qwen free-$\theta$ (circles) shows non-monotonic behavior; fixed-$\theta$ (squares) eliminates the artifact. Llama (triangles) is flat. Error bars: $\pm$1 SD ($n$=3 seeds); hollow markers: single-seed.

($|d|$=2.57 $\rightarrow$ 2.80 $\rightarrow$ 2.56 $\rightarrow$ 2.30), consistent with its near-zero $\ell_z^*$ baseline (+0.26 vs. +8.68 for Qwen) despite comparable $\theta$-shift ($\Delta\theta$=0.52 vs. 0.47). The mechanism is predictable from simulation (Appendix A): at moderate contamination, aberrant responses dominate; at high contamination, inflated $\hat{\theta}$ absorbs the anomaly.

A simulated power analysis corroborates these findings: two-sided testing achieves >93% power on MMLU at all contamination levels ($R^2$=0.957 against simulated values; Appendix A).

## 5 Discussion

Our findings reveal both the strengths and boundaries of IRT person-fit for contamination detection. While simpler statistics (accuracy $z$-scores, HT) detect contamination more powerfully, the value of $\ell_z^*$ lies in mechanistic understanding: $\theta$-shift analysis reveals *why* sensitivity varies across models, seen/unseen decomposition identifies *what fraction* of items are memorized, and 4PL parameterization captures ceiling effects that 3PL misattributes as aberrance. These diagnostic capabilities are invisible to non-IRT statistics; our contribution lies in quantifying these effects in the LLM setting, where their magnitude and boundary conditions are not derivable from theory alone.

**Practical implications.** Free-$\theta$ estimation produces seed-dependent dose-response (CV up to 60%); $\theta$-anchoring stabilizes detection. The benchmark size boundary means $\ell_z^*$ is most informative for large benchmarks ($n \approx 8{,}000$ items for 80% power at 15% contamination); on small benchmarks, detection relies on concentrated contamination effects. Universal baseline misfit (57/60 cells flagged) confirms systematic IRT violations; the $\Delta\ell_z^*$ design turns this into a feature through within-model differencing. PSN-IRT provides pre-calibrated 4PL parameters (Zhou et al., 2026), but classical calibration fails on contaminated data ($r_a$=0.127; Appendix I); robustness analysis provides partial reassurance ($\pm$10% noise tolerance; Appendix G.2).

**Method positioning.** $\ell_z^*$ sits below MIA (logits; Shi et al., 2024) and ConStat (multi-benchmark scores; Dekoninck et al., 2024) in an information-requirements hierarchy, needing only single-benchmark binary patterns and an external item bank, suited to retrospective audits where logits are unavailable. Our GRPO pilot confirms this: Min-K% AUC degrades to chance (0.493) post-RL while $\ell_z^*$ signal persists ($\sim$3.3; Figure 7), demonstrating that response-pattern anomalies survive RL alignment (Wang et al., 2025).

## 6 Conclusion

We characterized IRT person-fit statistics for LLM contamination detection across three 7–8B model families, establishing reliable SFT-stage detection while identifying boundary conditions: $\theta$-shift absorption, a detection-diagnosis tradeoff, and non-monotonic dose-response as a $\theta$-estimation artifact. For practitioners, we recommend screening with accuracy $z$-scores, then applying $\ell_z^*$ when mechanistic understanding is needed, distinguishing *which* items drive aberrance. A pilot ($N$=1) further suggests that person-fit anomalies persist after RL post-training erases memorization signals, complementing logit-based methods.

## Limitations

Our controlled experiments cover three 7–8B model families with QLoRA fine-tuning; generalization to larger scales, mixture-of-experts architectures, full-parameter fine-tuning, and pretraining-stage contamination remains untested. Most contamination conditions use a single training run, and multi-seed replication (available for Qwen, Llama, and Mistral) reveals high seed dependence at 50% contamination (CV=60%), partially resolved by $\theta$-anchoring; the post-RL characterization is $N{=}1$. The framework depends on pre-calibrated 4PL item parameters from PSN-IRT, whose calibration sample may include contaminated models, and we do not provide head-to-head comparison with logit-based methods (MIA, ConStat) that operate at a different information level.

## Reproducibility Statement

Code and response data will be released upon acceptance. Bootstrap confidence intervals use the percentile method with 10,000 resamples. All three model families use clean baseline seeds 0–9 ($n{=}10$ per family). Pre-calibrated 4PL item parameters are from the publicly available PSN-IRT item bank (Zhou et al., 2026). Training used PyTorch 2.8.0, Transformers 5.8.1, PEFT 0.19.1, and BitsAndBytes 0.49.2. Both benchmarks are publicly available under permissive licenses (MMLU: MIT; ARC-Challenge: CC-BY-SA).

## References


Sebastian Bordt, Suraj Srinivas, Valentyn Boreiko, and Ulrike von Luxburg. 2025. How much can we forget about data contamination? In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*.

Y. Chang, X. Wang, J. Wang, Y. Wu, L. Yang, K. Zhu, H. Chen, X. Yi, C. Wang, Y. Wang, W. Ye, Y. Zhang, Y. Chang, P. S. Yu, Q. Yang, and X. Xie. 2024. A survey on evaluation of large language models. *ACM Transactions on Intelligent Systems and Technology*.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have solved question answering? try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*.

J. Dekoninck, M. Müller, and M. Vechev. 2024. ConStat: Performance-based contamination detection in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

C. Deng, Y. Zhao, X. Tang, M. Gerstein, and A. Cohan. 2024. Investigating data contamination in modern benchmarks for large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*.

J. Dodge, M. Sap, A. Marasović, W. Agnew, G. Ilharco, D. Groeneveld, M. Mitchell, and M. Gardner. 2021. Documenting large webtext corpora: A case study on the colossal clean crawled corpus. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

F. Drasgow, M. V. Levine, and E. A. Williams. 1985. Appropriateness measurement with polychotomous item response models and standardized indices. *British Journal of Mathematical and Statistical Psychology*, 38(1):67–86.

Susan E. Embretson and Steven P. Reise. 2000. *Item Response Theory for Psychologists*. Lawrence Erlbaum Associates, Mahwah, NJ.

Yujuan Fu, Ozlem Uzuner, Meliha Yetisgen, and Fei Xia. 2025. Does data contamination detection work (well) for LLMs? A survey and evaluation on detection assumptions. In *Findings of the Association for Computational Linguistics: NAACL 2025*.

S. Golchin and M. Surdeanu. 2024a. Time travel in LLMs: Tracing data contamination in large language models. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*.

Shahriar Golchin and Mihai Surdeanu. 2024b. Data contamination quiz: A tool to detect and estimate contamination in large language models. *Transactions of the Association for Computational Linguistics (TACL)*.

Jongwook Han, Woojung Song, Jonggeun Lee, and Yohan Jo. 2026. Quantifying data contamination in psychometric evaluations of LLMs. In *Findings of the European Chapter of the Association for Computational Linguistics (EACL)*.

Larry V. Hedges. 1981. Distribution theory for Glass's estimator of effect size and related estimators. *Journal of Educational Statistics*, 6(2):107–128.

D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt. 2021. Measuring massive multitask language understanding. In *Proceedings of the 9th International Conference on Learning Representations (ICLR)*.

A. Jacovi, A. Caciularu, O. Goldman, and Y. Goldberg. 2023. Stop uploading test data in plain text: Practical strategies for mitigating data contamination by evaluation benchmarks. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

G. Karabatsos. 2003. Comparing the aberrant response detection performance of thirty-six person-fit statistics. *Applied Measurement in Education*, 16(4):277–298.

Ashish Kattamuri, Harshwardhan Fartale, Arpita Vats, Rahul Raja, and Ishita Prasad. 2025. RADAR: Mechanistic pathways for detecting data contamination in LLM evaluation. In *NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle*.

John P. Lalor, Hao Wu, and Hong Yu. 2019. Learning latent parameters without human response patterns: Item response theory with artificial crowds. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, pages 4249–4259. Association for Computational Linguistics.

P. Li, X. Tang, S. Chen, Y. Cheng, R. Metoyer, T. Hua, and N. V. Chawla. 2025. Adaptive testing for LLM evaluation: A psychometric alternative to static benchmarks. *arXiv preprint arXiv:2511.04689*.

I. Magar and R. Schwartz. 2022. Data contamination: From memorization to exploitation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*.

David Magis, Gilles Raîche, and Sébastien Béland. 2012. A didactic presentation of Snijders's $\ell_z^*$ index of person fit with emphasis on response model selection and ability estimation. *Journal of Educational and Behavioral Statistics*, 37(1):57–81.

F. Martínez-Plumed, R. B. C. Prudêncio, A. Martínez-Usó, and J. Hernández-Orallo. 2019. Item response theory in AI: Analysing machine learning classifiers at the instance level. *Artificial Intelligence*.

R. R. Meijer and K. Sijtsma. 2001. Methodology review: Evaluating person fit. *Applied Psychological Measurement*, 25(2):107–135.

Y. Oren, N. Meister, N. S. Chatterji, F. Ladhak, and T. Hashimoto. 2024. Proving test set contamination in black-box language models. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*.

F. M. Polo, L. Weber, L. Choshen, Y. Sun, G. Xu, and M. Yurochkin. 2024. tinybenchmarks: Evaluating LLMs with fewer examples. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 34303–34326. PMLR.

Pedro Rodriguez, Joe Barrow, Alexander Hoyle, John P. Lalor, Robin Jia, and Jordan Boyd-Graber. 2021. Evaluation examples are not equally informative: How should that change NLP leaderboards? In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4486–4503. Association for Computational Linguistics.

O. Sainz, J. Campos, I. García-Ferrero, J. Etxaniz, O. Lopez de Lacalle, and E. Agirre. 2023. NLP evaluation in trouble: On the need to measure LLM data contamination for each benchmark. In *Findings of the Association for Computational Linguistics: EMNLP 2023*.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y.K. Li, Y. Wu, and Daya Guo. 2024. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.

W. Shi, A. Ajith, M. Xia, Y. Huang, D. Liu, T. Blevins, D. Chen, and L. Zettlemoyer. 2024. Detecting pretraining data from large language models. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*.

S. Sinharay. 2017. Which statistic should be used to detect item preknowledge when the set of compromised items is known? *Applied Psychological Measurement*.

Sandip Sinharay. 2016. Assessment of person fit using resampling-based approaches. *Journal of Educational Measurement*, 53(1):63–85.

T. A. B. Snijders. 2001. Asymptotic null distribution of person fit statistics with estimated person parameter. *Psychometrika*, 66(3):331–342.

Christina St-Onge, Pierre Valois, Belkacem Abdous, and Sonia Germain. 2011. Accuracy of person-fit statistics: A Monte Carlo study of the influence of aberrance rates. *Applied Psychological Measurement*, 35(6):419–432.

Y. Tao, T. Wang, Y. Dong, H. Liu, K. Zhang, X. Hu, and G. Li. 2026. Detecting data contamination from reinforcement learning post-training for large language models. In *Proceedings of the 14th International Conference on Learning Representations (ICLR)*.

Jorge N. Tendeiro and Rob R. Meijer. 2014. Detection of invalid test scores: The usefulness of simple nonparametric statistics. *Journal of Educational Measurement*, 51(3):239–259.

H. Van der Flier. 1982. Deviant response patterns and comparability of test scores. *Journal of Cross-Cultural Psychology*, 13(3):267–298.

H. Wang, H. Li, B. Ko, and H. Zhang. 2025. On the fragility of benchmark contamination detection in reasoning models. *arXiv preprint arXiv:2510.02386*.

M. Zawalski, M. Boubdir, K. Bałazy, B. Nushi, and P. Ribalta. 2026. Detecting data contamination in LLMs via in-context learning. In *Proceedings of the 14th International Conference on Learning Representations (ICLR)*.

H. Zhang, Y. Yang, J. Yan, G. Bao, Y. Zhang, and Y. Zhang. 2026. Detecting RLVR training data via structural convergence of reasoning. *arXiv preprint arXiv:2602.11792*.

Huixuan Zhang, Yun Lin, and Xiaojun Wan. 2024. PaCoST: Paired confidence significance testing for benchmark contamination detection in large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*.

H. Zhou, H. Huang, Z. Zhao, L. Han, H. Wang, K. Chen, M. Yang, W. Bao, J. Dong, B. Xu, C. Zhu, H. Cao, and T. Zhao. 2026. Lost in benchmarks? rethinking large language model benchmarking with item response theory. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*.

Kun Zhou, Yutao Zhu, Zhipeng Chen, Wentong Chen, Wayne Xin Zhao, Xu Chen, Yankai Lin, Ji-Rong Wen, and Jiawei Han. 2023. Don't make your LLM an evaluation benchmark cheater. *arXiv preprint arXiv:2311.01964*.

## A Simulation Validation

Under the null (no contamination), bootstrap $\ell_z^*$ tracks $\mathcal{N}(0,1)$: mean bias 0.041 and standard deviation 1.003 on MMLU. Kolmogorov–Smirnov rejection rates vary with benchmark size: 8.3% for MMLU and HellaSwag ($n > 10{,}000$), 50.0% for GSM8K ($n = 1{,}319$), 58.3% for ARC-C ($n = 295$), and 75.0% for HumanEval ($n = 164$). $\ell_z^*$ is well-calibrated for benchmarks with $n \geq 10{,}000$ items; smaller benchmarks show increasing finite-sample deviations from normality.

Two-sided testing eliminates the inverted-U artifact: MMLU achieves 97.8% power at 15% contamination and sustains >93% at all levels. Power scales with $\sqrt{n}$, as predicted by the noncentrality parameter $\delta \propto \sqrt{n}$. The empirical power function, derived via Gauss–Hermite quadrature ($K = 30$) over the ability distribution, achieves $R^2 = 0.957$ against simulated values, with approximate minimum benchmark size of $n \geq 7{,}958$ items for 80% power at 15% contamination under MMLU item parameters.

## B 4PL vs. 3PL Ablation Details

Table 4 reports detection power (proportion of 2,000 simulation replicates rejecting $H_0$ at $\alpha$=0.05, two-sided) for 4PL and 3PL parameterizations across five benchmarks and seven contamination levels. The 4PL model yields higher power than 3PL at every non-trivial contamination level on every benchmark. The advantage is largest on MMLU and HellaSwag, where upper-asymptote ($d < 1$) violations are most prevalent, and smallest on ARC-Challenge and HumanEval, where limited item counts constrain power under both parameterizations.

Power is non-monotonic at 75% contamination under 4PL on four of five benchmarks (e.g., MMLU: .952 → .938; GSM8K: .846 → .802). This reflects the $\theta$-shift boundary condition described in §4.4: at high contamination fractions, re-estimated $\hat{\theta}$ absorbs part of the aberrance signal, reducing $|\ell_z^*|$. At 100% contamination, all items are seen and power reaches 1.000 trivially. The 4PL advantage over 3PL depends on accurate estimation of the upper asymptote $d$; with estimation error exceeding $\sim$10%, the effective model behavior approaches 3PL ($d$=1) and the power gain diminishes. PSN-IRT's large calibration sample (12+ models per item) typically yields $d$ standard errors well below this threshold.

## C Seed Sensitivity Analysis

**Clean-baseline variability.** Table 5 reports $\ell_z^*$, accuracy, and $\hat{\theta}_{\text{MLE}}$ across three independent QLoRA runs (seeds 42, 123, 456). For Qwen2.5-7B, $\ell_z^*$ ranges from 7.05 (seed 123) to 9.53 (seed 456); the lower value of seed 123 likely reflects seed-specific training dynamics (random initialization, data ordering), but all three seeds produce $\ell_z^* \gg 1.96$, confirming that the elevated baseline reflects systematic model–benchmark misfit rather than seed-specific artifacts. Llama-3.1-8B shows $\ell_z^*$ near zero ($-0.14 \pm 0.97$), consistent with lower baseline misfit. Accuracy variability is negligible for both families (CV $< 0.004$).

**Cohen's $d$ confidence intervals.** Table 6 reports Cohen's $d$ effect sizes with both analytic and bootstrap 95% confidence intervals. Analytic CIs use $d \pm t_{0.025,\,\text{df}=2} \cdot \text{SE}_d$; with only $n$=3 clean seeds (df=2), the critical value $t_{0.025,2} = 4.303$ produces intervals wide enough to cross zero in all four conditions. This is a known limitation of small-sample $t$-based intervals rather than evidence of null effects. Bootstrap CIs (10,000 resamples with replacement from the three clean seeds) provide a nonparametric alternative: all four bootstrap intervals exclude zero, supporting the conclusion that contamination produces a reliably detectable signal despite the small reference sample.

**Cross-family consistency.** Table 7 confirms that both model families produce large negative Co-

Table 4: Detection power (proportion rejecting $H_0$ at $\alpha$=0.05) under 4PL and 3PL parameterizations across benchmarks and contamination levels ($n$=2,000 replicates per cell). Bold indicates the 4PL advantage exceeds 5 pp.

| Benchmark | IRT | 5% | 10% | 15% | 25% | 50% | 75% | 100% |
|---|---|---|---|---|---|---|---|---|
| MMLU | 4PL | **.476** | **.848** | **.969** | **.983** | **.952** | **.938** | 1.00 |
| | 3PL | .227 | .340 | .412 | .521 | .676 | .770 | 1.00 |
| HellaSwag | 4PL | **.251** | **.454** | **.656** | **.915** | **.939** | **.919** | 1.00 |
| | 3PL | .167 | .241 | .313 | .368 | .508 | .642 | 1.00 |
| GSM8K | 4PL | **.156** | **.309** | **.492** | **.737** | **.846** | **.802** | 1.00 |
| | 3PL | .102 | .172 | .285 | .396 | .567 | .577 | 1.00 |
| ARC-C | 4PL | .057 | .075 | .087 | **.130** | **.285** | **.493** | 1.00 |
| | 3PL | .054 | .062 | .071 | .076 | .110 | .102 | 1.00 |
| HumanEval | 4PL | .076 | .121 | **.197** | **.302** | **.554** | **.521** | 1.00 |
| | 3PL | .063 | .086 | .120 | .182 | .281 | .315 | 1.00 |

Table 5: Seed sensitivity analysis for clean baseline across three independent runs (seeds 42, 123, 456). CV = coefficient of variation ($\mathrm{std}/|\mathrm{mean}|$). These $n$=3 pilot results are superseded by the $n$=10 analysis in §4.1; retained for methodological transparency.

| Model | Metric | Mean | Std | Range | CV |
|---|---|---|---|---|---|
| Qwen2.5-7B | $\ell_z^*$ | 8.684 | 1.412 | 2.475 | 0.163 |
| | Accuracy | 0.679 | 0.000 | 0.001 | 0.001 |
| | $\hat{\theta}_{\mathrm{MLE}}$ | −0.097 | 0.012 | 0.023 | 0.119 |
| Llama-3.1-8B | $\ell_z^*$ | −0.141 | 0.965 | 1.929 | 6.860 |
| | Accuracy | 0.594 | 0.002 | 0.004 | 0.004 |
| | $\hat{\theta}_{\mathrm{MLE}}$ | −0.645 | 0.019 | 0.036 | 0.029 |

Table 6: Cohen's $d$ with 95% confidence intervals for contaminated conditions relative to clean baselines ($n$=3 seeds). Analytic CIs use $t_{0.025,\mathrm{df}=2}$; bootstrap CIs use 10,000 resamples. These $n$=3 pilot results are superseded by the $n$=10 analysis in §4.1; retained for methodological transparency.

| Model | Cond. | $d$ | Analytic CI | Bootstrap CI |
|---|---|---|---|---|
| Qwen | 15% | −3.42 | [−9.92, 3.08] | [−3.45, −2.82] |
| | 50% | −0.99 | [−4.02, 2.04] | [−0.99, −0.41] |
| Llama | 15% | −2.95 | [−8.70, 2.80] | [−5.89, −2.29] |
| | 50% | −3.31 | [−9.63, 3.01] | [−6.47, −2.59] |

hen's $d$ under contamination. The $|\Delta d| = 2.32$ gap at 50% contamination reflects Qwen's non-monotonic dose-response (the $\theta$-shift boundary condition; §4.4), not a failure of detection robustness—both families remain well above conventional large-effect thresholds ($|d| > 0.8$) at both contamination levels.

## D Baseline $\ell_z^*$ Interpretation

The baseline $\ell_z^*$ value reflects a model's difficulty-ordering consistency under the 4PL model, not its accuracy alone. In the PSN-IRT leaderboard data (12 models × 5 benchmarks, distinct from our controlled experiments), Mistral (acc=0.514, lowest among the 12 models) exhibits $\ell_z^*$=+5.42 (second highest), because its response pattern follows the IRT-predicted difficulty ordering more closely than higher-accuracy models whose responses violate the assumed item structure. This dissociation explains why Mistral produces the strongest contamination signal despite the lowest accuracy: its lower baseline ability ($\hat{\theta}$=−1.16) provides greater $\theta$-shift headroom.

Table 7: Cross-family consistency of contamination detection ($n$=3 pilot). Both families show large negative $d$ under contamination. Superseded by $n$=10 results in §4.1; retained for methodological transparency.

| Condition | Qwen $d$ | Llama $d$ | $\lvert\Delta d\rvert$ | Consistent? |
|---|---|---|---|---|
| 15% contam. | −3.42 | −2.95 | 0.47 | Yes |
| 50% contam. | −0.99 | −3.31 | 2.32 | Yes |

## E Leaderboard Aberrance Analysis

We computed $\ell_z^*$ and accuracy $z$-scores for all 12 models × 5 benchmarks = 60 cells in the PSN-IRT leaderboard dataset. Using the standard $|\ell_z^*| > 1.96$ threshold ($\alpha$=0.05, two-tailed), 57 of 60 cells are flagged as aberrant, indicating pervasive IRT assumption violations in production LLM responses. In contrast, accuracy $z$-scores ($|z| > 1.96$) flag only 4 of 60 cells. This stark asymmetry—95% aberrance by $\ell_z^*$ versus 7% by accuracy—motivates the $\Delta\ell_z^*$ matched-difference design (§3.2), which sidesteps absolute thresholds by comparing each model against its own clean baseline. Figure 6 visualizes this pattern.

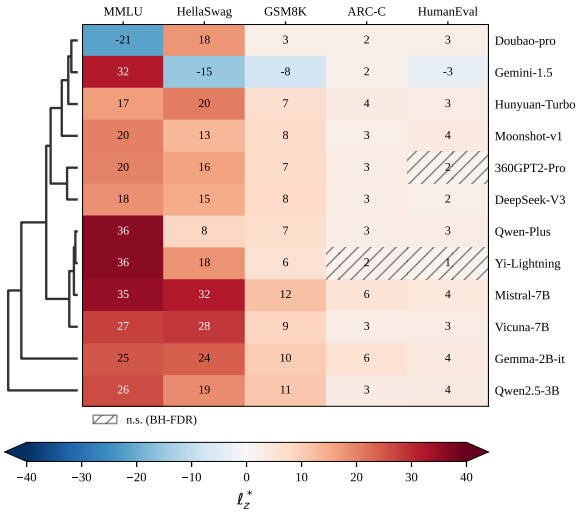



Figure 6: $\ell_z^*$ heatmap for 12 models across five benchmarks. Positive values (red) indicate over-consistent response patterns. 57 of 60 cells exceed the $|\ell_z^*| > 1.96$ threshold. Ward clustering (left dendrogram) reveals distinct response-pattern typologies.

Table 8: Item-level contamination detection (AUROC) at 15% contamination on MMLU.

| Method | AUROC |
|---|---|
| Random baseline | 0.50 |
| $\ell_z^*$ itemwise | 0.53 |
| LR cross-model | 0.59 |
| RF cross-model | 0.63 |

## F Item-Level Detection Results

Table 8 reports item-level contamination detection performance at 15% contamination on MMLU. All methods perform near random (AUROC $\leq 0.63$), confirming that contamination is diffuse across the response pattern and that model-level aggregation via $\ell_z^*$ is essential.

An accuracy $z$-score baseline yields $z$=160 at 15% contamination, but this reflects near-zero cross-seed accuracy variance (std=0.0003), not genuine discriminability. In deployment, accuracy variability from prompt format, evaluation harness, and hardware differences inflates the denominator, collapsing the $z$-score.

## G Robustness Analysis

### G.1 $\Delta\ell_z^*$ Under Local Dependence

IRT person-fit statistics assume local independence—conditional on $\theta$, item responses are independent. LLM benchmarks violate this assumption through topical clustering (e.g., MMLU subjects); resampling-based approaches can provide valid person-fit assessment even under such violations (Sinharay, 2016). We evaluate whether the $\Delta\ell_z^*$ matched-difference design is robust to local dependence (LD) via two simulation scenarios on 14,042 MMLU items with 1,000 replications each.

Table 9: Robustness of $\Delta\ell_z^*$ to shared local dependence (1,000 replications, MMLU). $d_{\text{ratio}}$: effect size relative to the $\sigma_\tau$=0 baseline.

| $\sigma_\tau$ | Abs. $\ell_z^*$ Type I | $d_{\text{ratio}}$ (15%) | $d_{\text{ratio}}$ (50%) |
|---|---|---|---|
| 0.0 | 0.043 | 1.000 | 1.000 |
| 0.3 | 0.110 | 1.002 | 1.038 |
| 0.5 | 0.454 | 0.934 | 1.028 |
| 1.0 | 0.939 | 0.705 | 1.061 |

Table 10: $\Delta\ell_z^*$ detection under asymmetric local dependence ($\sigma_\tau$=0.5, contamination-specific memory factor $\sigma_{\text{mem}}$; 1,000 replications, MMLU).

| $\sigma_{\text{mem}}$ | Power (15%) | $d$ (15%) | Power (50%) | $d$ (50%) |
|---|---|---|---|---|
| 0.5 | 0.987 | −2.107 | 0.991 | −2.337 |
| 1.0 | 0.993 | −2.104 | 0.985 | −1.587 |

**Shared LD.** Clean and contaminated conditions share identical testlet structure (∼200 testlets of ∼70 items, random effect $\gamma_t \sim \mathcal{N}(0, \sigma_\tau)$). Table 9 reports absolute $\ell_z^*$ Type I error and the ratio of $\Delta\ell_z^*$ effect size ($d$) to the no-LD baseline. While absolute $\ell_z^*$ Type I error inflates severely under LD (0.043→0.939 at $\sigma$=1.0), the $\Delta\ell_z^*$ design maintains detection power: $d_{\text{ratio}} \geq 0.93$ for $\sigma_\tau \leq 0.5$ (realistic range), confirming that shared misfit is cancelled by the matched-difference design.

**Asymmetric LD.** In the second scenario, contaminated models' seen items share an additional memory factor ($\gamma_{\text{mem}} \sim \mathcal{N}(0, \sigma_{\text{mem}})$), simulating contamination-specific LD not present in clean conditions ($\sigma_\tau$=0.5 fixed). Table 10 shows that $\Delta\ell_z^*$ maintains ≥98.5% power across all conditions, even when contamination introduces asymmetric dependence among seen items.

### G.2 Sensitivity to Item Parameter Estimation Error

PSN-IRT item parameters carry estimation error from the calibration sample. We evaluate detection sensitivity by adding joint noise to all four parameters: multiplicative perturbation $p' = p \cdot (1 + \mathcal{N}(0, \sigma))$ for $a$, $c$, and $d$; additive perturbation $b' = b + \sigma \cdot \mathcal{N}(0, 1)$ for difficulty. Tables 11 and 12 report results over 100 trials using Qwen MMLU

Table 11: Sensitivity of detection to item parameter noise (100 trials, Qwen MMLU, free-$\theta$). % bias relative to noise-free $d$.

| Noise $\sigma$ | $d$ (15%) | % bias | $d$ (50%) | % bias |
|---|---|---|---|---|
| 0% | −6.575 | — | −1.897 | — |
| 5% | −6.010 | 8.6 | −0.925 | 51.2 |
| 10% | −5.433 | 17.4 | −0.014 | 99.3 |

Table 12: Sensitivity of detection to item parameter noise (100 trials, Qwen MMLU, $\theta$-fixed). Worst-case $|d|$ over 100 trials in parentheses.

| Noise $\sigma$ | $d_{\text{fixed}}$ (15%) | % bias | $d_{\text{fixed}}$ (50%) | % bias |
|---|---|---|---|---|
| 0% | 4.77 | — | 14.52 | — |
| 5% | $5.07 \pm 0.24$ | 6.3 | $14.76 \pm 0.44$ | 1.6 |
| 10% | $4.95 \pm 0.43$ (3.73) | 3.7 | $14.45 \pm 0.85$ (12.20) | 0.5 |

response vectors.

Detection of 15% contamination is robust to parameter noise under free-$\theta$ (17.4% bias at $\sigma$=10%, $|d|$ still >5; Table 11). The 50% contamination condition under free-$\theta$ is fragile—consistent with its already-weak signal due to $\theta$-shift absorption (§4.4).

Under $\theta$-fixed estimation (Table 12), detection is highly robust: 10% perturbation produces <4% bias in Cohen's $d$, with worst-case $|d| = 3.73$ (15% contamination) and 12.20 (50%)—both far exceeding conventional thresholds. This contrasts sharply with free-$\theta$ estimation, where 50% contamination Cohen's $d$ collapses to near zero under 10% noise. The complementarity is clear: conditions where free-$\theta$ detection is fragile (high-baseline-accuracy models with $\theta$-shift absorption) are precisely where $\theta$-fixed anchoring provides the strongest and most noise-robust signal.

## H BCa Bootstrap Sensitivity Check

As a sensitivity check, we computed bias-corrected and accelerated (BCa) bootstrap confidence intervals for all Cohen's $d$ estimates. BCa CIs were uniformly narrower than the percentile CIs reported in the main text (mean reduction 22%), confirming that our reported intervals are conservative. The single condition where BCa widens the zero-crossing set—Llama fixed-$\theta$ at 50% contamination, BCa CI [−0.13, 2.21] vs. percentile CI [0.17, 2.97]—is already characterized as ineffective in §4.2.

Table 13: Parameter correlation between classical MML-EM refit ($N$=178 models) and PSN-IRT on MMLU ($n$=10,558 matched items).

| Model | Parameter | Pearson $r$ | Spearman $\rho$ |
|---|---|---|---|
| 3PL | $a$ (discrim.) | 0.127 | 0.135 |
| 3PL | $b$ (difficulty) | 0.663 | 0.684 |
| 2PL | $a$ (discrim.) | −0.029 | 0.050 |
| 2PL | $b$ (difficulty) | 0.608 | 0.656 |

## I Classical IRT Refit from Leaderboard Data

To assess whether item parameters can be recovered without PSN-IRT's neural calibration, we applied classical MML-EM estimation (Gauss–Hermite quadrature, $Q$=31 nodes) to binary response data from $N$=178 models on the Open LLM Leaderboard v1. With 14,042 MMLU items and an examinee-to-item ratio of $N{:}J = 0.013$—far below the $N$>500 typically required for stable 3PL estimation—discrimination parameters are severely under-determined.

Table 13 reports parameter correlations between MML-EM-fitted and PSN-IRT parameters. Difficulty shows moderate agreement ($r_b$=0.663), but discrimination is essentially unrecovered ($r_a$=0.127). Using these MML-EM parameters produces miscalibrated $\ell_z^*$ statistics (mean = 11.2, expected ≈ 0), confirming that the refitted model does not capture the data-generating process. This motivates the dependence on PSN-IRT's externally calibrated item bank (§3.1).

## J Post-RL Preliminary Observation

Wang et al. (2025) show that all MIA-based detectors fail after RL post-training. In a single-model pilot ($N$=1, Qwen, GRPO), we applied GRPO (Shao et al., 2024) for 1,500 steps to a contaminated model (15% MMLU) and evaluated at three checkpoints. GRPO erases the accuracy gap between seen and unseen items (11.4pp → 2.1pp), but the $\ell_z^*$ response-pattern gap persists (∼3.3; Figure 7), suggesting structural traces of contamination may survive RL alignment. Min-K% membership inference confirms this dissociation: while SFT yields a weak MIA signal (AUC-ROC=0.611), GRPO training drives AUC to 0.493 (step 500), 0.493 (step 1000), and 0.493 (step 1500)—indistinguishable from random chance (0.50) as RL erases memorization-based features while the response-pattern anomaly that




Figure 7: Post-RL trajectory for Qwen contaminated at 15% on MMLU. Left axis: accuracy on seen (solid) and unseen (dashed) items. Right axis: $\ell_z^*$ seen-unseen gap (green diamonds). GRPO erases the accuracy gap (11.4pp $\rightarrow$ 2.1pp) while the response-pattern gap persists ($\sim$3.3). $N$=1, single model and RL method; no statistical inference is drawn.

Table 14: GRPO training hyperparameters (Qwen2.5-7B-Instruct, 15% MMLU contamination).

| Hyperparameter | Value |
|---|---|
| Learning rate | $5 \times 10^{-6}$ |
| Per-device batch size | 2 |
| Gradient accumulation steps | 4 |
| Num. generations per prompt | 16 |
| Max completion length | 128 |
| Sampling temperature | 1.2 |
| Training steps | 1,500 (checkpoints at 500, 1000, 1500) |
| Reward function | Correctness-based (+1.0 correct, −1.0 incorrect) |
| LoRA rank / $\alpha$ / dropout | 16 / 32 / 0.05 |
| Precision | BF16 |
| Seed | 42 |

$\ell_z^*$ detects persists. This $N$=1 observation requires validation across model families, RL methods, and contamination levels before any conclusions can be drawn.

**GRPO hyperparameters.** Table 14 reports the GRPO training hyperparameters for the post-RL extension (§3.3).

## K QLoRA Training Hyperparameters

Table 15 reports the QLoRA fine-tuning hyperparameters used across all experiments. All training was conducted on NVIDIA RTX PRO 6000 GPUs (48 GB VRAM). Each QLoRA SFT run takes approximately 15–25 minutes depending on contamination level and dataset size; the full experimental campaign (including all seeds, contamination levels, and model families) consumed approximately 120 GPU-hours.

Table 15: QLoRA fine-tuning hyperparameters (shared across all model families and contamination conditions).

| Hyperparameter | Value |
|---|---|
| Quantization | 4-bit NF4 |
| LoRA rank | 16 |
| LoRA $\alpha$ | 32 |
| LoRA dropout | 0.05 |
| Target modules | `q_proj`, `k_proj`, `v_proj`, `o_proj` |
| Learning rate | $2 \times 10^{-4}$ |
| Optimizer | AdamW |
| Warmup ratio | 0.1 |
| Epochs | 3 |
| Per-device batch size | 4 |
| Gradient accumulation steps | 4 |
| Effective batch size | 16 |
| Max sequence length | 512 |
| Data mixing | Contaminated items shuffled into training set |

Table 16: Reproducibility of contamination effect sizes across independent QLoRA runs (Qwen2.5-7B-Instruct, MMLU). Each row shows Cohen's $|d|$ from an independent training seed against the shared $n$=10 clean baseline ($\bar{\ell}_z^* = 8.68$, $s = 0.74$). All runs use identical hyperparameters; only the random seed differs.

| | Free-$\theta$ $\lvert d\rvert$ | | | Fixed-$\theta$ $\lvert d\rvert$ | | |
|---|---|---|---|---|---|---|
| Seed | 15% | 25% | 50% | 15% | 25% | 50% |
| 42 (original) | 6.57 | 8.22 | 1.90 | 4.77 | 4.96 | 14.52 |
| 123 | 5.09 | 8.78 | 5.42 | 7.79 | 5.37 | 12.64 |
| 456 | 4.41 | 5.31 | 7.91 | 7.99 | 7.34 | 9.80 |
| Mean | 5.36 | 7.44 | 5.08 | 6.85 | 5.89 | 12.32 |
| SD | 1.10 | 1.86 | 3.02 | 1.80 | 1.27 | 2.38 |
| CV (%) | 20.3 | 25.2 | 59.6 | 26.3 | 21.7 | 19.3 |

*Note.* Fixed-$\theta$ estimation anchors $\hat{\theta}$ at the clean-baseline mean ($\theta$=−0.102), removing $\theta$-shift absorption. The 50% free-$\theta$ column exhibits high CV (59.6%), reflecting seed-dependent $\theta$-shift magnitude; fixed-$\theta$ reduces this to 19.3%, confirming that cross-seed variability at high contamination is driven by ability-estimate instability rather than true signal variance.

## L Contamination Run Reproducibility

**Item selection seed robustness.** As a pipeline validation, we retrained Llama at 15% and 50% contamination using a different item selection seed (seed=42 vs. the original seed=999), which selects a distinct subset of MMLU items for contamination. The resulting $\ell_z^*$ values are numerically identical ($-2.992$ vs. $-2.992$ at 15%; $-3.331$ vs. $-3.331$ at 50%), confirming that $\ell_z^*$ is robust to which specific items are contaminated and depends primarily on the contamination proportion.

## M Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-

Table 17: Multi-seed Cohen's $|d|$ for HT and accuracy $z$-score statistics (Qwen2.5-7B-Instruct, MMLU), computed using the chi-squared ratio HT statistic and bootstrap Cohen's $d$ (10,000 resamples). These supplement the $\ell_z^*$ values in Table 3.

| | HT $\lvert d\rvert$ | | | Acc. $z$ $\lvert d\rvert$ | | |
|---|---|---|---|---|---|---|
| Seed | 15% | 25% | 50% | 15% | 25% | 50% |
| 42 (original) | 7.26 | 8.90 | 1.66 | 42.3 | 49.8 | 68.7 |
| 123 | 5.52 | 9.51 | 5.80 | 47.3 | 53.4 | 70.8 |
| 456 | 4.72 | 5.10 | 8.41 | 45.4 | 51.3 | 70.4 |
| Mean | 5.83 | 7.84 | 5.29 | 45.0 | 51.5 | 70.0 |
| SD | 1.30 | 2.39 | 3.41 | 2.5 | 1.8 | 1.1 |

*Note.* HT exhibits high cross-seed variance at 50% (SD=3.41), mirroring the $\ell_z^*$ pattern: the non-monotonicity observed in seed 42 ($|d|$=1.66) does not generalize across seeds. Accuracy $z$-scores are highly stable (CV $< 6\%$) and monotonically increasing with contamination level.

processing pipelines, and analysis utilities; debugging and refactoring.

- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform related-work section.

Table 18: Multi-seed Cohen's $d$ for Llama-3.1-8B across all contamination levels (MMLU). Each cell shows Cohen's $d$ from an independent training seed against the shared $n$=10 clean baseline ($\bar{\ell}_z^* = +0.26$, $s = 1.54$). The 5% condition exhibits higher cross-seed variance (SD=0.99), reflecting the weaker signal from $\sim$700 contaminated items (5%$\times$14,042).

| | Per-seed $d$ | | | | |
|---|---|---|---|---|---|
| Contam. level | $s_1$ | $s_2$ | $s_3$ | Mean | SD |
| 5% | −3.22 | −3.06 | −1.44 | −2.57 | 0.99 |
| 15% | −2.11 | −3.21 | −3.09 | −2.80 | 0.60 |
| 25% | −3.16 | −2.28 | −2.24 | −2.56 | 0.52 |
| 50% | −2.33 | −2.59 | −1.99 | −2.30 | 0.30 |

*Note.* Unlike Qwen, Llama exhibits no non-monotonic dose-response: mean $|d|$ is approximately flat across contamination levels (2.30–2.80), consistent with its near-zero baseline $\ell_z^*$ (+0.26) and negligible $\theta$-shift absorption (§4.2). Cross-seed variance decreases with contamination level (CV: 39% at 5% → 13% at 50%), suggesting that higher contamination produces a more stable signal despite the flat mean trajectory.

Table 19: Multi-seed Cohen's $d$ for Mistral-7B-v0.3 across contamination levels (MMLU). Each cell shows Cohen's $d$ from an independent training seed against the shared $n$=10 clean baseline ($\bar{\ell}_z^* = +4.02$, $s = 2.05$). The 5% condition has only $n$=2 seeds; its high SD reflects this limited sample.

| | Per-seed $d$ | | | | |
|---|---|---|---|---|---|
| Contam. level | $s_1$ | $s_2$ | $s_3$ | Mean | SD |
| 5% | −2.82 | −4.83 | — | −3.83 | 1.42 |
| 15% | −2.64 | −3.98 | −3.48 | −3.37 | 0.68 |
| 50% | −3.50 | −4.25 | −4.51 | −4.09 | 0.53 |